Problem Statement:

A client's requirement is, he wants to predict whether the person has the chronic kidney disease based on several parameters.

Domain: **Machine Learning**

Learning: **Supervised**

Type: **Classification**

Name of the Project: **Chronic Kidney Disease prediction**

Dataset:

Number of Records: **399**

Number of Columns: **25**

Column Names: **'age', 'bp', 'sg', 'al', 'su', 'rbc', 'pc', 'pcc', 'ba', 'bgr', 'bu', 'sc', 'sod', 'pot', 'hrmo', 'pcv', 'wc', 'rc', 'htn', 'dm', 'cad', 'appet', 'pe', 'ane', 'classification'**

Independent Variables: **'age', 'bp', 'sg', 'al', 'su', 'rbc', 'pc', 'pcc', 'ba', 'bgr', 'bu', 'sc', 'sod', 'pot', 'hrmo', 'pcv', 'wc', 'rc', 'htn', 'dm', 'cad', 'appet', 'pe', 'ane',**

Dependent Variable: **'classification'**

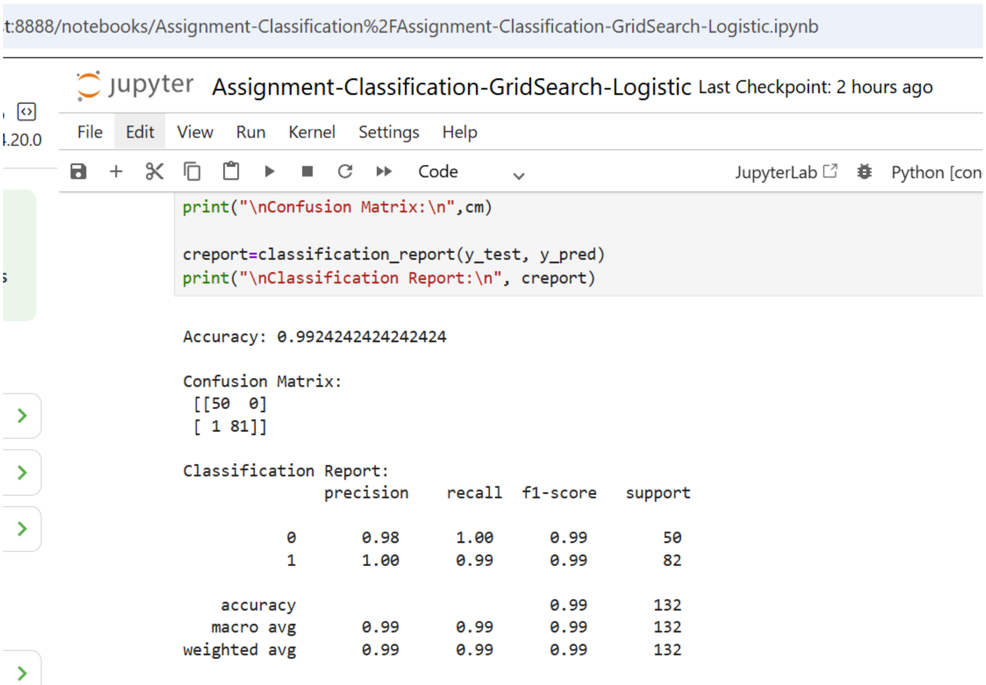
Ordinal data

Feature	**Values**
sg (Specific Gravity)	a, b, c, d, e
al (Albumin)	0, 1, 2, 3, 4, 5
su (Sugar)	0, 1, 2, 3, 4, 5
appet (Appetite)	poor < good

Nominal data

Feature	Values
rbc	normal, abnormal
pc	normal, abnormal
pcc	present, notpresent
ba	present, notpresent
htn	yes, no
dm	yes, no
cad	yes, no
pe	yes, no
ane	yes, no

GridSearch-Logistic




```
print("\nConfusion Matrix:\n",cm)

creport=classification_report(y_test, y_pred)
print("\nClassification Report:\n", creport)
```

```
Accuracy: 0.9924242424242424

Confusion Matrix:
 [[50  0]
 [ 1 81]]

Classification Report:
               precision    recall  f1-score   support

           0       0.98      1.00      0.99        50
           1       1.00      0.99      0.99        82

    accuracy                           0.99       132
   macro avg       0.99      0.99      0.99       132
weighted avg       0.99      0.99      0.99       132
```

Classification-GridSearch-SVM

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Code JupyterLab

```
accuracy=accuracy_score(y_test, y_pred)
print("\nAccuracy:", accuracy)

cm=confusion_matrix(y_test, y_pred)
print("\nConfusion Matrix:\n",cm)

creport=classification_report(y_test, y_pred)
print("\nClassification Report:\n", creport)
```

```
Accuracy: 0.9772727272727273

Confusion Matrix:
 [[49  1]
 [ 2 80]]

Classification Report:
              precision    recall  f1-score   support

           0       0.96      0.98      0.97        50
           1       0.99      0.98      0.98        82

    accuracy                           0.98       132
   macro avg       0.97      0.98      0.98       132
weighted avg       0.98      0.98      0.98       132
```

[ ]:

GridSearch-DecisionTree

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Code JupyterLab Python [conda env:base]

]
20.0

```
print("\nConfusion Matrix:\n",cm)

creport=classification_report(y_test, y_pred)
print("\nClassification Report:\n", creport)
```

```
Accuracy: 0.9696969696969697

Confusion Matrix:
 [[48  2]
 [ 2 80]]

Classification Report:
              precision    recall  f1-score   support

           0       0.96      0.96      0.96        50
           1       0.98      0.98      0.98        82

    accuracy                           0.97       132
   macro avg       0.97      0.97      0.97       132
weighted avg       0.97      0.97      0.97       132
```

[ ]:

Classification-GridSearch-RandomForest

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Code JupyterLab Python [conda env:base] *

```
Best Parameters:
{'classifier__class_weight': None, 'classifier__criterion': 'entropy', 'classifier__max_depth':
ax_features': 'sqrt', 'classifier__n_estimators': 20}
```

[3]:

```python
##Printing Accuracy , Confusion Matric and Classification Report
from sklearn.metrics import accuracy_score, classification_report, confusion_matrix
accuracy=accuracy_score(y_test, y_pred)
print("\nAccuracy:", accuracy)

cm=confusion_matrix(y_test, y_pred)
print("\nConfusion Matrix:\n",cm)

creport=classification_report(y_test, y_pred)
print("\nClassification Report:\n", creport)
```

```
Accuracy: 1.0

Confusion Matrix:
 [[50  0]
 [ 0 82]]

Classification Report:
              precision    recall  f1-score   support

           0       1.00      1.00      1.00        50
           1       1.00      1.00      1.00        82

    accuracy                           1.00       132
   macro avg       1.00      1.00      1.00       132
weighted avg       1.00      1.00      1.00       132
```

NaviesBayes

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Code JupyterLab Python [conda env:base] * Anaconda

```
y_pred = best_model.predict(X_test)

Best Parameters:
{'classifier__var_smoothing': 1e-09}
```

[3]:

```python
##Printing Accuracy , Confusion Matric and Classification Report
from sklearn.metrics import accuracy_score, classification_report, confusion_matrix
accuracy=accuracy_score(y_test, y_pred)
print("\nAccuracy:", accuracy)

cm=confusion_matrix(y_test, y_pred)
print("\nConfusion Matrix:\n",cm)

creport=classification_report(y_test, y_pred)
print("\nClassification Report:\n", creport)
```

```
Accuracy: 0.9924242424242424

Confusion Matrix:
 [[50  0]
 [ 1 81]]

Classification Report:
              precision    recall  f1-score   support

           0       0.98      1.00      0.99        50
           1       1.00      0.99      0.99        82

    accuracy                           0.99       132
   macro avg       0.99      0.99      0.99       132
weighted avg       0.99      0.99      0.99       132
```

K-Nearest Neighbour

:8888/notebooks/Assignment-Classification%2FAssignment-Classification-KNN.ipynb

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Code JupyterLab Python [conda env:l

```
creport=classification_report(y_test, y_pred)
print("\nClassification Report:\n", creport)
```

```
Accuracy: 0.9848484848484849

Confusion Matrix:
 [[50  0]
 [ 2 80]]

Classification Report:
               precision    recall  f1-score   support

           0       0.96      1.00      0.98        50
           1       1.00      0.98      0.99        82

    accuracy                           0.98       132
   macro avg       0.98      0.99      0.98       132
weighted avg       0.99      0.98      0.98       132
```

Algorithm	Accuracy
GridSearch-Logistic	0.9924
GridSearch-SVM	0.9772
GridSearch-DecisionTree	0.9696
Classification-GridSearch-RandomForest	1
NaviesBayes	0.9924
K-Nearest Neighbour	0.9848